* Discrete Mathematics *

Discrete means :- Natural numbers are discrete N = 1, 2, 3, ... and real numbers are not discrete.

Discrete Mathmetics :- Discrete mathmatics is the subject of discrete objects.

Example :- Every finite set is discrete A = {0, 1, 2}

Advantage of discrete mathmatics :-

1. It developers your mathmatical thinking.
2. If you want to make a career in computer science then you should study it.

Set

(~~set~~ well define unordered collection of distinct element is called a set)

A set is the mathmatical model for a collection of different things; a set contains elements or numbers. which can be mathmatical objects of any kind: numbers, symbols, points in space, lines, other geometrical shapes, variables, or even other sets.

Ex

A = {a, e, i, o, u}

Types of set :- [i] Empty set :- A set having no element is called a empty set it is donated by ϕ or { }

[ii] ~~Singleton set :- A set having no elements is called an empty set it is denote.~~

Singleton set :- A set having only one element is called singleton set

Subset :- A set is called a subset of the set B if for all $x \in A \Rightarrow x \in B$, it is expressed as $A \subseteq B$

Ex $B = \{a, e, i, o, u\}$
$A = \{a, e, i\}$, then $A \subseteq B$

Power set :- The family of all subsets of a set A is called the power set of A.

Ex $A = \{a, b\}$, $P(A) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

FUNDAMENTAL OPERATION OF SET :-

Union	$A \cup B$	$\{x \mid x \in A$ or $x \in B\}$
Intersection	$A \cap B$	$\{x \mid x \in A$ & $x \in B\}$
Difference	$A - B$	$\{x \mid x \in A$ & $x \notin B\}$

CARTESIAN PRODUCT OF TWO SETS :-

Let A & B are two sets then $A \times B = \{(x, y) \mid x \in A, y \in B\}$

Example Let $A = \{1, 2\}$ and $B = \{3, 4\}$

Then $A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$

Relation :- Let A and B are two sets then every relation from A to B is a subset of $A \times B$.

$\{(1, 3) (1, 4)\}$

$\{(1, 3) ; (4, 2), (2, 3)\}$

formula: $2^{m \times n}$

Q-1. If $A=\{1,2\}$, $B=\{2,3\}$, $C=\{3,7\}$ Then determine $(A\times B)\cap(A\times C)$.

$A\times B=\{(1,2)(1,3),(2,2),(2,3)\}$

$A\times C=\{(1,3),(1,7),(2,3),(2,7)\}$

$(A\times B)\cap(A\times C)=\{(1,3),(2,3)\}$ //

Q-2 If A,B,C,D are any four sets then prove that

$(A\cap B)\times(C\cap D)=(A\times C)\cap(B\times D)$

Ans L.H.S

$(A\cap B)\times(C\cap B)\Rightarrow(x,y)\in(A\cap B)\times(C\cap D)$

$\Rightarrow x\in(A\cap B)\times y\in(C\cap D)$

$\Rightarrow(x\in A\ \&\ x\in B)\ \&\ (y\in C\ \&\ y\in D)$

$\Rightarrow(x\in A\times y\in C)\ \&\ (x\in B\times y\in D)$

$\Rightarrow(x,y)\in(A\times C)\ \&\ (x,y)\in(B\times D)$

$\Rightarrow(x,y)\in(A\times C)\cap(B\times D)$

$\Rightarrow(A\cap B)\times(C\cap D)\subseteq(A\times C)\cap(B\times D)$

$(A\times C)\cap(B\times D)\Rightarrow(x,y)\in(A\times C)\cap(B\times D)$

$\Rightarrow x\in(A\times C)\ \&\ y\in(B\times D)$

$\Rightarrow(x\in A\ \&\ x\in C)\ \&\ (y\in B\ \&\ y\times D)$

$\Rightarrow(x\in y)\in(A\cap B)\times(x\in y)\in(C\cap D)$

$\Rightarrow(x\in y)\in(A\cap B)\times(C\cap D)$

$\Rightarrow(A\times C)\cap(B\times D)\subseteq(A\cap B)\times(C\cap D)$ //

Principal of mathematical induction

Suppose P is a proposition defined on the positive integer N and suppose P has the following two properties.

(A) P(1) is true.

(B) P(k+1) is true when whenever P(k) is true. Then P is true for every ~~integer~~ positive integer.

Q-1 show by mathematical induction

$$1^2+3^2+5^2+\cdots+(2n-1)^2 = \frac{n(2n+1)(2n-1)}{3}$$

A for n=1,

$$P(1) \Rightarrow 1^2 = \frac{1(2\times1+1)(2\times1+1)}{3} = \frac{1\times3\times1}{3} = 1$$

∴ This is true for n=1.

Now let it P true for n=k

$$P(k) \Rightarrow 1^2+3^2+5^2+\cdots+(2k-1)^2 = \frac{k(2k+1)(2k-1)}{3}$$

Now PROVE for n=k+1

$$1^2+3^2+5^2+\cdots+(2(k+1)-1)^2 = \frac{(k+1)[2(k+1)+1][2(k+1)-1]}{3}$$

$$= \frac{(k+1)(2k+3)(2k+1)}{3}$$

$$1^2+3^2+5^2+\cdots+(2k+1)^2+(2k+1)^2 = \frac{k(2k+1)(2k-1)}{3} + (2k+1)^2$$

$$= (2k+1)\left[\frac{k(2k-1)}{3} + (2k+1)\right]$$

$$= (2k+1)\left[\frac{2k^2-k+6k+3}{3}\right]$$

$$= \frac{(2k+1)(2k+3)(2k+1)}{3}$$

//

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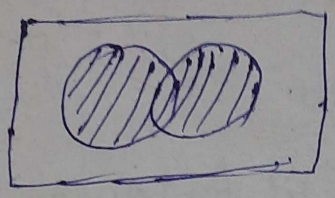
VENN Diagram

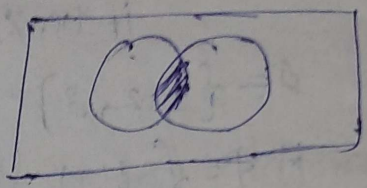
Pictorial representation of a set by area written circles with elements representation by point written them are called their Venn Diagram.

Let U be a universal set and A, B are two sets.

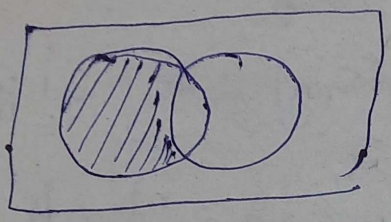
(1) $A \cup B$



(2)

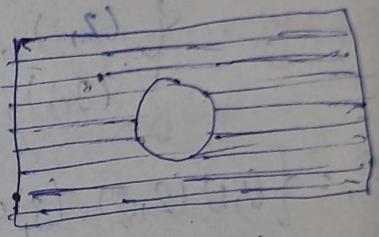


(3)



$A - B$

(4) A^c



C = complement

Set identities

Let's consider some basic set identities.
Let say A, B and C are three sets and these three sets are subsets of the universal set U.

(1) identity laws

(i) $A \cup \phi = A$

(ii) $A \cap U = A$

(i) $A \cup \phi = A$

$A \cup \phi = \{x \mid x \in A \vee x \in \phi\}$

We know that $x \in A$ is false

$A \cup \phi = \{x \mid x \in A \vee f$

$= \{x \mid x \in A$

$A \cup \phi = A$

(ii) $A \cap U = A$

$A \cap U = \{x \mid x \in A \wedge x \in U$

$\because x \in U$ is always true

$= \{x \mid x \in A \wedge T$

$= \{x \mid x \in A$

$= A$ //

(2) Domination Laws

(i) $A \cup U = U$

(ii) $A \cap \phi = \phi$

(3) Idempotent Laws:

(i) $A \cup A = A$

(ii) $A \cap A = A$

Reflexive Relation: A Relation 'R' on a set 'A' is said to be Reflexive if $(x, x) \in R \; \forall \; x \in A$

$A = \{1, 2, 3\}$

$A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$R_1 = \{(1,1) (2,2) (3,3)\}$ — True

$R_2 = \{(1,1), (2,2)\}$ — False

$R_3 = \{(1,1) (2,2) (3,3) (1,2)\}$ — True

Irreflexive Relation: A Relation R on a set 'A' is said to be irreflexive if $x \not R x$ {x is not related to x} $\forall \; x \in A$

Ex $A = \{1, 2, 3\}$

$A \times A = \{(1,1) (1,2) (1,3) (2,1) (2,2) (2,3) (3,1) (3,2) (3,3)\}$

$R = \{(1,1) (1,2)\}$ — false

$R = \{(1,2) (1,3) (2,1)\}$ — True

$R = \{(1,1) (2,2)\}$ — false

Symmetric Relation :- A Relation R on a set 'A' is said to be symmetric if xRy then $yRx \forall x,y \in A$.

i.e if $(x,y) \in R$ than $(y,x) \in R \forall (x,y) \in A$

$A = \{1,2,3,4\}$

$R_1 = \{(1,2)(1,3)(2,1)(2,2)(3,1)\}$ — True

$R_2 = \{\}$ — True

Antisymmetric : A Relation 'R' is said to be antisymmetric if xRy and yRx then $x=y \forall x,y \in A$ i.e.

if $(x,y) \in R$ then $(y,x) \in R$ only if $(x=y) \forall x,y \in A$.

Ex $A = \{1,2,3\}$

$R = \{(1,2)(2,2)(2,1)\}$ # It is symmetric but not Antisymmetric.

$R = \{(1,1)(2,2)(1,3)\}$ # it is Antisymmetric.

Asymmetric :- A Relation 'R' is said to be asymmetric if xRy and yRx then $x=y \forall x,y \in A$.

i.e if $(x,y) \in R$ then $(y,x) \in R \forall x,y \in A$

$A = \{1,2,3\}$

$R_1 = \{(1,2)(1,3)(1,1)\}$ # False

$R_2 = \{(1,2)(1,3)(2,3)\}$ # True

$R_3 = \{\}$ — True

Note

(i) Every Asymmetric is antisymmetric — True

(ii) Every Antisymmetric is asymmetric — False

Theorems proof

1) Proof that if $(A \subseteq B$ and $B \subseteq C)$, $A \subseteq C$

Ans $\because$

$A \subseteq B = \{x / x \in A \Rightarrow x \in B$

$B \subseteq C = \{x / x \in B \Rightarrow x \subseteq C$

similiarly

$\{x / x \in A \Rightarrow x \Rightarrow C = A \subseteq C$ //

(2) Proof that $A \subseteq B$ and $B \subseteq C$, $C \subseteq A$ then $A \subseteq C$

Ans if $A \subseteq B$ & $B \subseteq C$

then

$A \subseteq C$

given $C \subseteq A$

if $A \subseteq C$ & $C \subseteq A$

similiarly $A = C$

Given

$A \subseteq B = \{x / x \in A \Rightarrow x \subseteq B$

$B \subseteq C = \{x / x \in B \Rightarrow x \in C$

$\therefore A \subseteq C$

similiarly

$C \subseteq A$

if $A \subseteq C$ & $C \subseteq A$

hence, $A = C$

Q. Write down power set of

(i) ϕ

Ans We know that

$\phi = \{\}$

$P(\phi) = \{\phi\}$

Proved

(ii) $\{\phi, \{\phi\}\}$

Let $A = \{\phi, \{\phi\}\}$

$P(A) = \{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$

Union of set:- Union of set A and B is defined to be the set of all those elements which belong to 'A or B' or both and it's denoted by 'A∪B'

$\therefore A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Q $A = \{1, 2, 3, 4, 5\}$

$B = \{1, 3, 7, 0\}$

$A \cup B = \{1, 2, 3, 4, 5, 7, 0\}$

Intersection of sets:- Intersection of two sets A and B is the set of all those elements which belong to both 'A and B' and it denoted by 'A∩B'

$\therefore A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

$A = \{1, 2, 3, 5, 9, 6\}$

$B = \{2, 6, 4\}$

$A \cap B = \{2, 6\}$

Difference of sets:- Difference of 2 set A and B is a set of all those elements which belongs to A but do not belong to set B.

$A - B = \{x \mid x \in A \text{ and } x \notin B\}$

Q $A = \{2, 3, 9, 5\}$

$B = \{2, 1, 0, 9, 7\}$

$A - B = \{3, 9, 5\}$

1 Complement of a set with respect to Universal set:- The complement of a set A is a set of all those element of universal set which do not belong to A and is denoted by A^c or A'.

$A' = \cup A = \{x | x \in U \text{ and } x \notin A\}$

$= \{x | x \notin A\}$

Symmetric Difference of sets

The symmetric difference of two sets A and B is the set containing all elements that are in A or in B but not in both and is denoted by.

$A \oplus B$.

$A \oplus B = (A \cup B) - (A \cap B)$ or $A \oplus B = (A - B) \cup (B - A)$

Ex

$A = \{a, b, c, d, e\}$

$B = \{a, b, h, i, j\}$

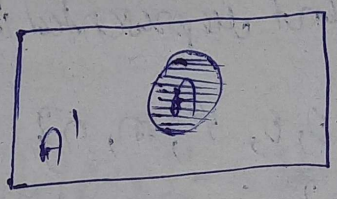
$A \oplus B = \{c, d, e, h, i, j\}$

Laws	UNION	Intersection (n)
Idempotent laws	$A \cup A = A$	$A \cap A = A$
Associative laws	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive laws	$A \cup (B \cap C) = (A \cap B) \cap (A \cup C)$	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
commutative laws	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Demorgen Laws	$(A \cup B)' = A' \cap B'$	$(A \cap B)' = A' \cup B'$
Identity laws	$A \cup \phi = A$ & $A \cup U = U$	$A \cap \phi = \phi$ & $A \cap \phi = A$
Complement laws	$A \cup A' = U$ & $U' = \phi$	$A \cap A' = \phi$ & $\phi' = U$
Absorption laws	$A \cup (A \cup B) = A$	$A \cap (A \cup B) = A$
Involution laws	$(A')' = A$	

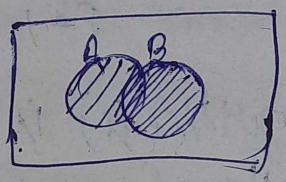
Venn Diagram represents sets as sets operations as regions in place :-

→ All sets are usually represented by circle.
→ Universal set is represented by a rectangle.
→ The complement of the set is represented by that portion of the universal set which is not in the set.

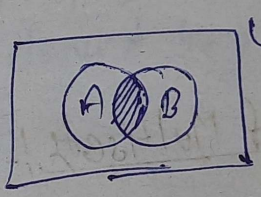
(i) $A \cup A$



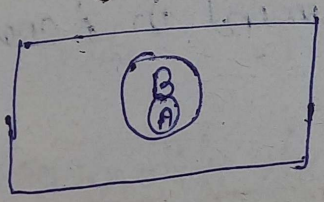
(ii) $A \cup B$



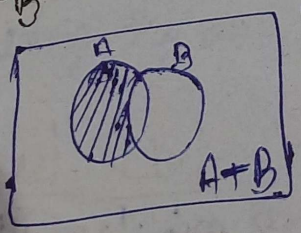
(iii) $A \cap B$



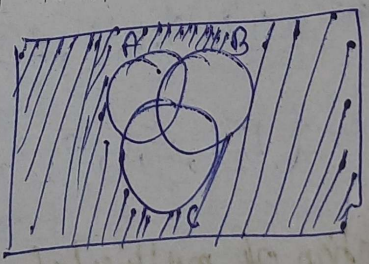
(iii) $A \subset B$



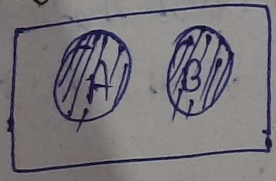
(iv) $A - B$



(v) $A' \cap B' \cap C'$



(vi) $A \cup B$



1 Set of sets:- A set can have its elements as another set.

Eg $A = \{a, \{a\}, \{a, b, c\}\}$

Multisets

It is an unordered collection of element in which the multiplicity of an element may be one or more than one or zero.

Multiplicity → No of times element represented in set.

Q (i) $A = \{l, l, l, m, n, o, o, p, m, n\}$ — $l=3$, $m=2$, $n=3$, $o=2$, $p=1$

(ii) $B = \{a, a, a, a, a, c\}$

① Union of Multisets:- $A \cup B$ (A & B are multisets) is a multiset such that multiplicity of an element is equal to maximum of the multiplicity of an element in A and B.

Eg $A = \{1, 1, 2, 2, 2, 3, 3, 3, 3\}$

$B = \{1, 2, 2, 2, 2, 2, 2, 3, 3\}$

$A \cup B = \{1, 1, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3\}$

(ii) Intersection of multisets:- The intersection of 2 ~~minute~~ multisets A and B is a multiset such that the multiplicity of the element is ~~A and~~ B minimum of the multiplicity of the element in A and B denoted by $A \cap B$.

$A = \{1, 1, 2, 3, 5, 6, 6, 6\}$

$B = \{1, 2, 4, 6\}$

$A \cap B = \{1, 2, 6\}$ //

(iii) Difference of Multisets:- It is multiset such that that the multiplicity of an element is equal to the multiplicity of the element in A minus the multiplicity of the element in B, and only the positive are equal to zero the considered.

Ex $A = \{1, 1, 1, 2, 2, 3, 2, 5\}$

$B = \{1, 1, 2, 2, 2, 4, 4, 3, 3, 3, 3, 3, 5, 5, 5\}$

$A - B = \{1\}$ //

(iv) Sum of Multisets (A+B):- It is the multiset such that the multiplicity of an element is equal to the sum of Multiplicity of the element in A and B.

Ex $A = \{1, 3, 4, 4, 5, 6\}$

$B = \{1, 1, 3, 3, 4, 5, 6, 7\}$

$A+B = \{1, 1, 3, 3, 3, 4, 4, 4, 5, 5, 5, 6, 7\}$ //

(v) Cardinality of Multisets:- It is number of distinct elements in a multiset without considering the multiplicity of en element.

Ordered Pair:- An ordered pair consists of two elements such that one of them is designed as first member and other as second member. (P, Q)
Let p is first & q is second element then ordered pair is (P, Q)

Ex $A = \{1, 2\}$ then possible ordered pair $A \times A$

$A \times A = \{(1, 1) (1, 2) (2, 1) (2, 2)\}$

1. Cartesian product:- Cartesian product of two set P and Q is the set of all ordered pairs where 1st number is from set P & 2nd number is from set Q & it is denoted by $P \times Q$.

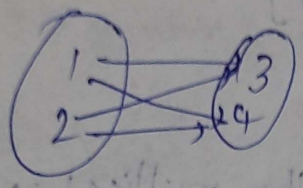
$$P \times Q = \{(x, y), x \in P \text{ and } y \in Q\}$$

$$P \times Q \neq Q \times P$$

Ex

$P = \{1, 2\}$

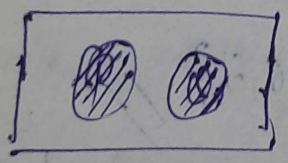
$Q = \{3, 4\}$



$P \times Q = \{(1,3)\ (1,4)\ (2,3)\ (2,4)\}$

2. Principle of inclusion & exclusion

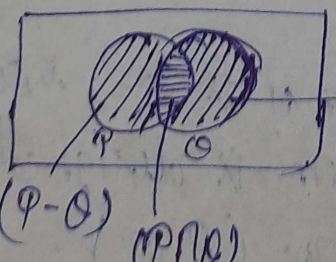
First principle:- If P and Q are two disjoint set.



$P \cap Q = \phi$

$$|P \cup Q| = |P| + |Q|$$

Second If P and Q are 2 non disjoint set than:-



$$|P \cup Q| = |P - Q| + |Q - P| + |P \cap Q| \quad \text{---(1)}$$

$$|Q| = |Q - P| + |P \cap Q| \quad \text{---(2)}$$

$$|P| = |P - Q| + |P \cap Q| \quad \text{---(2)}$$

Adding both equation -

$|P| + |Q| = |P-Q| + |Q-P| + |P \cap Q| + |P \cap Q|$

$|P| + |Q| = |P \cup B| + |P \cap Q|$

$\boxed{|P \cup Q| = |P| + |Q| - |P \cap Q|}$

Similiarly :-

Then for $|P \cup Q \cup R|$, we can write it as $|P \cup (Q \cup R)|$

$|P \cup (Q \cup R)| = |P| + |Q \cup R| - |P \cap (Q \cup R)|$

$= |P| + |Q| + |R| - |Q \cap R| - |P \cap (Q \cup R)|$ — (1)

As we know

$P \cap (Q \cup R) = (P \cap Q) \cup (P \cap R)$ by definition of distributive law.

$|P \cap (Q \cup R)| = |P \cap Q| + (P \cap R) - |P \cap Q \cap R|$ — (2)

$\boxed{|P \cup (Q \cup R)| = |P| + |Q| + |R| - |Q \cap R| - |P \cap Q| - |P \cap R| + |P \cap Q \cap R|}$

Inclusion Principle in general for finite sets :-

$\boxed{|P_1 \cup P_2 \cup \dots P_n| = \sum |P_i| + \sum |P_i \cap P_j| + \sum |P_i \cap P_j \cap P_k| + (-1)^{n-1} |P_1 \cap P_2 \cap P_n|}$

Relations (R)

Relation :- A relation R from a set $A \times B$ is subset of $A \times B$.

$\boxed{R \subseteq A \times B}$

Q⇒ How many maximum Relation subset is posible from $A \times B$ and B where $|A| = m$, $|B| = n$.

Ans $|A| = m$, $|B| = n$

$A \times B = m \cdot n$

$\boxed{R_{m \cdot n} (A \times B) = 2^{m \cdot n}}$

Question ⇒ 1. Among 100 students, 32 study mathematics. 20 study physics. 45 study biology. 15 study mathematics and biology. 7 study mathematics and physics and 10 student study physics and biology and 30 do not study any of the three subjects.

[i] Find the number of students studying all three subjects.

[ii] Find the number of students studying exactly one of the three subjects.

Ans ⇒ Let M, P and B denote the number of students who study math., physics & biology.

$n(M) = 32$
$n(P) = 20$
$n(B) = 45$
$n(M \cap P) = 7$
$n(M \cap B) = 15$
$n(P \cap B) = 10$

$\therefore n(M \cup P \cup B) = 70$

(i) $n(M \cap P \cap B) = ?$

By inclusion and exclusion -

$$n(M \cup P \cup B) = n(M) + n(P) + n(B) - n(M \cap P) - n(M \cap P) - n(P \cap B) + n(M \cap P \cap B)$$

$$70 = 32 + 20 + 45 - 7 - 15 - 10 + n(M \cap P \cap B)$$

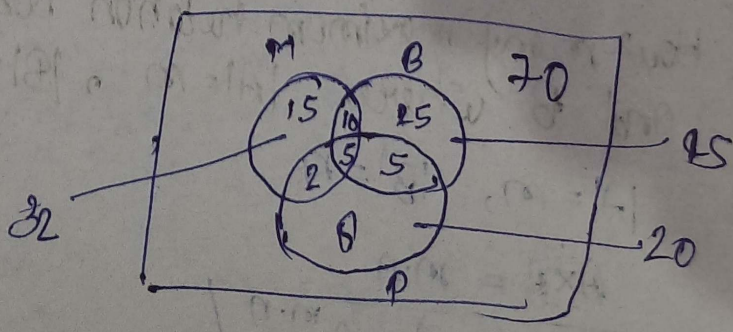
$$70 = 65 + n(M \cap P \cap B)$$

$$\boxed{n(M \cap P \cap B) = 5}$$

(ii).

Studying exactly one of the three subject $= 15 + 8 + 25$

$= 48$ //



Q=12 Out of 120 students surveyed it was found that 20 student have studied French. 50 students have studied English. 70 students have studied Hindi, 5 have studied English and French. 20 have studied English and Hindi, 10 have studied Hindi and French. Only 3 students have studied all the three languages. Find how many students have studied (i) Hindi alone (ii) French alone (iii) English but not Hindi (iv) Hindi but not French.

Ans Let F, E and H denote the number of students who study in French, English and Hindi.

$n(F)=20$ $n(E\cap F)=5$

$n(E)=50$ $n(E\cap H)=20$

$n(H)=70$ $n(H\cap F)=10$

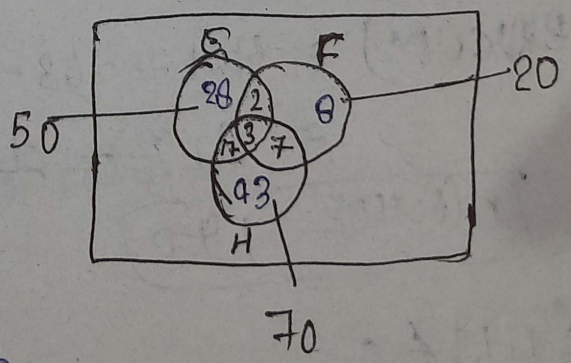
$n(F\cap E\cap H)=3$

$n(F\cup E\cup H)=n(F)+n(E)+n(H)-n(E\cap F)-n(E\cap H)-n(H\cap F)+n(F\cap E\cap H)$

$n(F\cup E\cup H)=20+50+70-5-20-10+3$

$=143-35$

$=108$ //



(i) Hindi alone-

$n(H)=70-17-3-7$

$=70-27$

$n(H)=43$

(ii) French alone

$n(F)=20-2-3-7$

$=20-12$

$=8$ //

(iii) $n(E\setminus H)=28+2=30$

(iv) $n(H\setminus F)=43+17=60$

Q. In a group of 191 students 40 are taking English, computer science and music, 36 are taking English and computer science, 20 are taking English and computer science, 20 are taking English and music, 18 are taking English and music, 18 are taking computer science and music, 10 are taking 65 are taking English, 76 are taking computer science and 63 are taking music.

Then answer the following -

(i) How many are taking English and music but not computer science.

(ii) How many are taking computer science and music but not computer science English.

(iii) How many are taking none of three subjects.

Ans ⇒ Let E, C and M is the number of student who get English, computer science and Music subject.

$n(E) = 65$ $n(E \cap C \cap M) = 10$

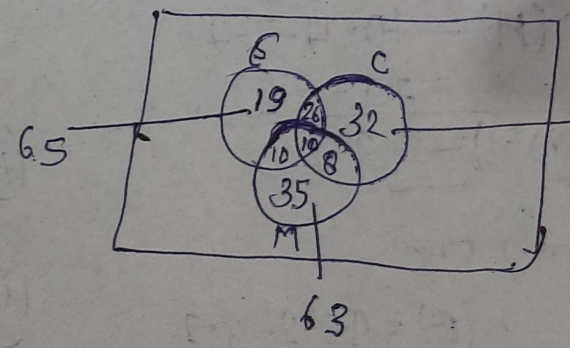
$n(C) = 76$ $n(E \cap C) = 36$ $n(C \cap M) = 18$

$n(M) = 63$ $n(E \cap M) = 20$

$n(E \cup C \cup M) = 65 + 76 + 63 - 36 - 20 - 18 + 10$

$= 214 - 24$

$n(E \cup C \cup M) = 190$



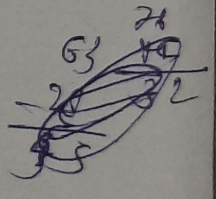
(i) $E \cap M \not\subset C$

$n(E \cap M \cap C') = 10$ //

(ii) $(C \cap M \cap E') = 8$ //

(iii) none of three subject $= 191 - 190$

$= 51$



* Partition set *

A partition of set A is a collection of non-empty subsets $A_1, A_2, A_3 \dots A_n$, called block, having these properties.

(1) A is the union of all the subsets. i-e

$$A_1 \cup A_2 \cup A_3 \dots \cup A_n = A$$

(2) The subsets are pairwise disjoint, $A_i \cap A_j = \phi$ for $i \neq j$
It means intersection of any two distinct subjects is empty.

Ex $A = \{1, 2, 3, 4, 5\}$ having a subset $B = \{2, 3\}$
$C = \{1, 4\}$ and $D = \{5\}$

Step I $A = \{1, 2, 3, 4, 5\}$

$B \cup C \cup D = \{1, 2, 3, 4, 5\}$

$B \cup C \cup D = A$

Step => II

$B \cap C = \phi$

$C \cap D = \phi$

$B \cap D = \phi$

$B \cap C \cap D = \phi$

All the subject of A is partition of set A.

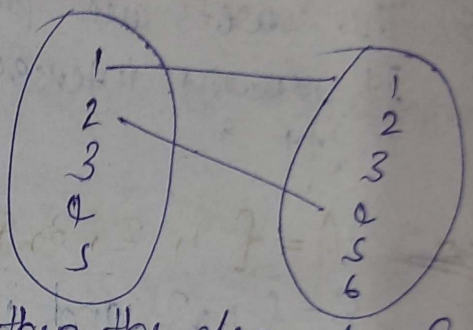
1. **Relation :-** Let A & B be two sets and some elements of A be in certain correspondance with some elements of B then that correspondance defines a relation between those elements of two sets.

(Or) ⇒ Let $A = \{1, 2, 3, 4 \dots\}$, $B = \{1, 2, 3, 4, \dots\}$

Let a relation R from A to B be define by $y = x$ where $y \in B$ & $x \in A$

i.e $R = \{(x, y) \; x \in A, y \in B \text{ & } y = x^2\}$

$R = \{(1,1) \; (2,4), (3,9) \dots\}$



Domain :- Let R be a relation of A to B then the element of A which are in relation are said to be domain.

RANGE :- Let R be a relation of A to B then the element of B which are in relation are said to be Range.

Ex If $R = \{(1,p), (1,q), (2,q)\}$

Domain $= \{1, 2\}$

Range $= \{p, q\}$

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Properties of Relation :-

1). **Reflexive relation :-** A relation R is said to reflexive if for all $x \in A$, $(x, x) \in A$

Ex Let $A = \{1, 2\}$ & $B = \{1, 2, 3\}$

$R = \{(1,1) \; (2,2), (1,3)\}$

Transitivity of a relation :- A relation R is transitive if $(x,y) \in R$ & $(y,z) \in R$, then $(x,z) \in R$.

Ex The relation "less than or equal to ($\leq$)" i.e.
$1 \leq 2$ & $2 \leq 4$ then $1 \leq 4$

Equivalence Relation :-
A relation R is called equivalence relation in A, if
(1) R is reflexive
(2) R is symmetric
(3) R is transitive.

Ex Question

1. Let $A = \{2,3,4\}$ and $B = \{3,4,5,7,6\}$ Assume a relation R from A to B such that $(x,y) \in R$ when x divides y. determine R, its domain & Range.

Ans The relation R is -
$R = \{(2,4), (2,6), (3,3) (3,6) (4,4)\}$

Domain $= \{2,3,4\}$

Range $= \{4,6,3\}$

2. Show that the relation
$R = \{(a,b) \mid a,b \in Z$ & $a-b$ is divisible by $3\}$ is an equivalence relation.

Ans Reflexive : for all $a \in Z$, $a-a=0$ is divisible by 3
Thus $(a,a) \in R$

(2) Symmetric : Let $(a,b) \in R$ then $a-b$ is divisibility by 3
$\Rightarrow -(a-b)$ is divisible by 3
$\Rightarrow (b-a)$ is divisible by 3
$\Rightarrow (b,a) \in R$

1. (iii) Transitive-

Let $(a,b) \in R$ & $(b,c) \in R$

$\Rightarrow a-b$ is divisible by 3 and $b-c$ is divisible by 3

$\Rightarrow a-b+b-c$ is divisible by 3

$\Rightarrow a-c$ is divisible by 3

$\Rightarrow (a,c) \in R$

So, R is an equivalence relation.

(2) If R is a relation on the set of integers such that $(a,b) \in R$ if $3a+4b = 7n$ for some integer n. Prove that R is an equivalence relation.

Sol (i) Reflexive: $3a+4a = 7a$ where $a \in Z$

$\Rightarrow (a,a) \in R$

(ii) Symmetric: if $(a,b) \in R$

$\Rightarrow 3a+4b = 7n$

Now for $3b+4a = 7b-4b+7a-3a$

$= 7b+7a-3a+4b$

$= 7(b+a-n)$

$= (b,a) \in R$

(iii) Transitive-

If $(a,b) \in R$ & $(b,c) \in R$

$\Rightarrow 3a+4b = 7n$ & $3b+4c = 7m$

Now, $3a+4b+3b+4c = 7m+7n$

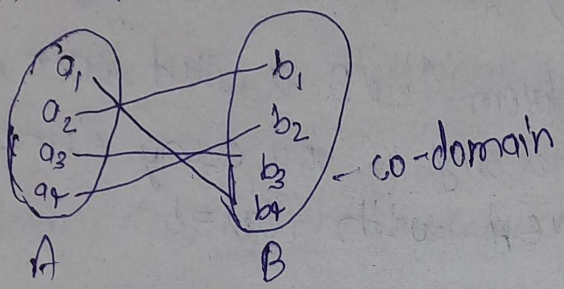
$3a+4c = 7(m+n-b)$

$(a,c) \in R$

Hence R is an equivalence relation.

* Functions and their types *

Function :- Link b/w two sets or b/w elements of the set is called as function or mapping. consider two sets 'A' and 'B' if for all $a \in A$, there exists a unique $b \in B$ such that $(a,b) \in f$



$f(a) = b \; \forall \; a \in A$ and $b \in B$

Types of Mapping :-

① **One-One (1-1) Mapping :-** It is also called as injective and it is (1-1) mapping. if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b.

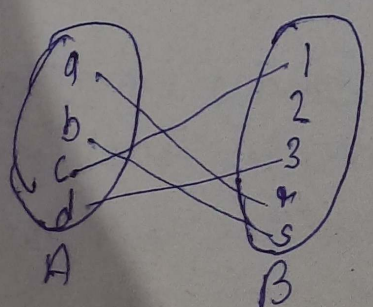
$a_1, a_2 \in A$

$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$

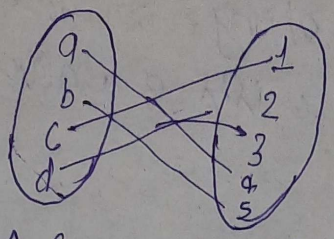
or if $a_1 \neq a_2, \Rightarrow f(a_1) \neq f(a_2)$

$A \to B$ is injective if all members of A are related to different members of B.

Q $\{a,b,c,d\}$ to $\{1,2,3,4,5\}$ $f(a)=4$, $f(b)=5$, $f(c)=1$ and $f(d)=3$.



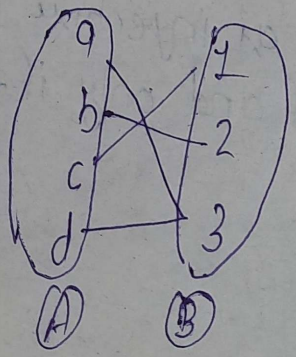
Ques:1 Determine whether the function f from {a,b,c,d} to {1,2,3,4,5} with f(a)=4, f(b)=5, f(c)=1 and f(d)=3 is one-to-one.



(ii) Onto Mapping:- A function f from A to B is called onto or Subjective if and only if for every element $b \in B$ - there is an element $a \in A$ with $f(a)=b$.

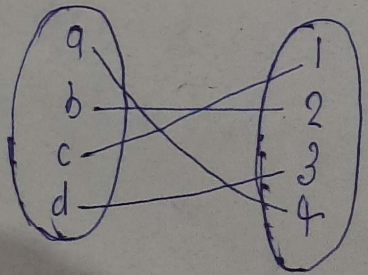
Ex

{a,b,c,d} to {1,2,3} defined by f(a)=3, f(b)=2, f(c)=1, f(a)=3, if f an onto function.



(iii) Bijective Mapping:- It is also called as one-to-one correspondance, if it is both one-one and onto.

Ques:1 Let f be the function from {a,b,c,d} to {1,2,3,4} with f(a)=4, f(b)=2, f(c)=1 and f(d)=3 is f a bijection?



* Recurrence Relation *

A recurrance relation of the sequence $\{a_n\}$ is an equation that expresses a_n in term of one or more of the previous terms of the sequence, namely $a_0, a_1, a_2, \ldots a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is non-negative integer.
A recurrence relations are also called difference equations.

Solution of Recurrence Relation: A sequence is called a solution of a recurrance relation of its terms satisfy the recurrence relation.

Ex Let $\{a_n\}$ be a sequence that satisfies that recurrence relation $a_n = a_{n-1} - a_{n-2}$, for $n = 2, 3, \ldots$ and suppose that $a_0 = 3$ and $a_1 = 5$ what are a_2 and a_3?

Solution $a_n = a_{n-1} - a_{n-2}$

given as

$a_0 = 3$

$a_1 = 5$

similarly -

$a_2 = a_{2-1} - a_{2-2} = a_1 - a_0$

$a_2 = 5 - 3 = 2$

$a_3 = a_{3-1} - a_{3-2} = a_2 - a_1$

$= 2 - 5 = -3$ //

* Linear Recurrence Relation with constant coefficients *

A recurrence relation of the form

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = f(n)$$

where c_i's are constant, is called a linear recurrence relation with constant coefficients.
The recurrence relation in equation above equation is known as a k^{th}-order recurrence relation, provided that both $c_0 \neq 0$ and $c_k \neq 0$

1. Second order Recurrence relation

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} = f(n)$$

Third order Recurrence relation

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} = f(n)$$

The solution of recurrence relation –

$$a_n = a_n^{(h)} + a_n^{(p)}$$

where $a_n = a_n^{(h)} + a_n^{(p)}$

$a_n^{(h)}$ = Homogeneous solution

$a_n^{(p)}$ = Particular solution

Homogeneous Recurrence Relation :- If $f(n) = 0$, then equation is called Homogeneous recurrence relation.

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = 0$$

Non-homogeneous Recurrence Relation :- If $f(n) \neq 0$, then given sequence is called non-homogeneous recurrence relation.

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = f(n)$$

Homogeneous linear Recurrence Relation with constant coefficients

Suppose the second order Homogeneous linear recurrence relation is

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} = 0$$

The characteristic Equation (Auxiliary equation) is

$$c_0 m^2 + c_1 m + c_2 = 0$$

Case-1 If the roots of A.E are real and unequal.

Say, $m_1 \neq m_2$

The general solution is

$$a_n = Cm_1^n + Cm_2^n$$

Case-2 If the roots of A.E are real and equal.

Say, $m_1 = m_2$

general solution -

$$a_n = (C_1 + nC_2)\lambda^n$$

Case-3 If the roots of A.E are in complex numbers

Say, $m = \alpha + i\beta$

general solution -

$$a_n = (C\cos n\theta + C_2 \sin n\theta)\, R^n$$

where

$$R = \sqrt{\alpha^2 + \beta^2}$$

and $\theta = \tan^{-1}\left(\frac{\beta}{\alpha}\right)$.

Problem-1 Solve the recurrence relation

$$a_n + 5a_{n-1} + 6a_{n-2} = 0$$

Ans given recurrence relation -

$$a_n + 5a_{n-1} + 6a_{n-2} = 0$$

characteristic Equation is

$$a_n = m^n$$

$$\therefore m^n + 5m^{n-1} + 6m^{n-2} = 0$$

divided by lowest power of m -

$$\frac{m}{m^{n-2}} + \frac{5m^{n-1}}{m^{n-2}} + \frac{6m^{n-2}}{m^{n-2}} = 0$$

$$m^2 + 5m + 6 = 0$$

$n^2 + 3n + 2n + 6 = 0$

$n(n+3) + 2(n+3) = 0$

$(n+3)(n+2) = 0$

$n = -2, -3$

general solution -

$$a_n = c_1(-2)^n + c_2(-3)^n$$

Problem-2 Solve the recurrence relation $a_n - 7a_{n-1} + 10a_{n-2} = 0$ given that $a_0 = 0, a_1 = 3$

Sol: given recurrence relation -

$$a_n - 7a_{n-1} + 10a_{n-2} = 0$$

characteristic equation -

$$m^2 - 7m + 10 = 0$$

$$m^2 - 5m - 2m + 10 = 0$$

$$m(m-5) - 2(m-5) = 0$$

$$(m-5)(m-2) = 0$$

$m = 5, 2$

general solution -

$$a_n = c_1 2^n + c_2 5^n \quad \text{---(i)}$$

$\because a_0 = c_1 \cdot 2^0 + c_2 \cdot 5^0$

$a_0 = c_1 + c_2$ ---(ii)

$\because a_1 = c_1 \times 2^1 + c_2 5^1$

$a_1 = 2c_1 + 5c_2$ ---(iii)

$0 = 2c_1 + 2c_2$

$3 = 3c_2$

$c_2 = 1$

putting in equation (i)

$c_1 + 1 = 0$

$c_1 = -1$

putting in equation (i)

$a_n = (-1)2^n + 1 \cdot 5^n$

$$a_n = (-1)2^n + 5^n$$

Problem - 3

Solve the recurrence relation

$a_n - 3a_{n-1} + 3a_{n-2} - a_{n-3} = 0$ given that $a_0 = 1$, $a_1 = -2$ and $a_2 = -1$

Sol given recurrence relation -

$$a_n - 3a_{n-1} + 3a_{n-2} - a_{n-3} = 0$$

characteristic equation -

$$m^3 - 3m^2 + 3m - 1 = 0$$

$\because (a-b)^3 = a^3 - b^3 + 3ab(a-b)$

$$(m-1)^3 = 0$$

$$m = 1, 1, 1$$

general solution -

$$a_n = (c_1 + c_2 n + c_3 n^2)(1)^n \quad \text{---(I)}$$

initial indexing -

$\because a_0 = 1$

$$c_1 + c_2 \times 0 + c_3 \times 0 = 1$$

$$c_1 = 1 \quad \text{---(II)}$$

$\because a_1 = -2$

$$c_1 + c_2 \times 1 + c_3 \times 1 = -2$$

$$c_1 + c_2 + c_3 = -2$$

$$1 + c_2 + c_3 = -2$$

$$c_2 + c_3 = -3 \quad \text{---(iii)}$$

$$c_2 = -1/2$$

$\because a_2 = -1$

$$c_1 + 2c_2 + 4c_3 = -1$$

$$1 + 2(c_2 + 2c_3) = -1$$

$$2(c_2 + 2c_3) = -2$$

$$c_2 + 2c_3 = -1 \quad \text{---(IV)}$$

$$c_3 = 5/2$$

putting in equation (I)

$$a_n = \left(1 + (-1/2)n + 5/2\, n^2\right)$$

1. Problem-4 Solve the recurrence relation relation.
$a_n + 2a_{n-1} + 2a_{n-2} = 0$ given that $a_0 = 0$, $a_1 = -1$

Ans

given recurrence relation-

$$a_n + 2a_{n-1} + 2a_{n-2} = 0$$

characteristic equation

$$m^2 + 2m + 2 = 0$$

solve by Shridharacharya-

$$am^2 + bm + c = 0$$

$$a = 1, \quad b = 2, \quad c = 2$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{4 - 4 \times 1 \times 2}}{2 \times 1}$$

$$m = \frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$m = \frac{-2 \pm \sqrt{-4}}{2} = \frac{-2 \pm \sqrt{2}i}{2}$$

$$m = -1 \pm i$$

$$m = \alpha + i\beta$$

$\therefore \quad \alpha = -1, \quad \beta = 1$

$$R = \sqrt{\alpha^2 + \beta^2} = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$$

$$R = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{-1}{1}\right) = \pi - \tan^{-1}(1)$$

$$\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

general equation-

$$a_n = \left[C_1 \cos\left(\frac{3\pi}{4} n\right) + C_2 \sin\left(\frac{3\pi}{4} n\right)\right] (\sqrt{2})^n \quad -(2)$$

Non-Homogeneous linear Recurrence Relation

Suppose the k^{th}-order Non-Homogeneous linear recurrence relation with constant coefficients is

$$c_0 a_n + c_1 a_{n-1} + c_2 a_{n-2} + c_3 a_{n-3} + \cdots + c_k a_{n-k} = f(n) \quad (i)$$

The solution of equation (i) is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

Case-I When $f(n) = b^n$ and b is not a root of characteristic equation.

Suppose the trial solution is $a_n = A \cdot b^n$

Q) Solve the recurrence relation -

$$a_n - 5a_{n-1} + 6a_{n-2} = 5^n$$

Ans A given recurrence relation -

$$a_n - 5a_{n-1} + 6a_{n-2} = 5^n$$

characteristic equation -

$$a_n = m^n$$

$$m^2 - 5m + 6 = 5^n$$

$$m^2 - 3m - 2m + 6 = 0$$

$$m(m-3) - 2(m-3) = 0$$

$$(m-3)(m-2) = 0$$

$$m = 2, 3$$

Homogeneous solution -

$$a_n^{(h)} = c_1 (2)^n + c_2 (3)^n$$

since $f(n) = 5^n$ and 5 is not root of homogeneous solution. then we will assume the particular solution is -

$$a_n^{(p)} = A \cdot (5)^n \quad \text{---(2)}$$

putting in equation (1)

$$A \cdot 5^n - 5(A \cdot 5^{n-1}) + 6(A \cdot 5^{n-2}) = 5^n$$

$$A \cdot 5^n \left(1 - 1 + \frac{6}{25}\right) = 5^n$$

$$A \cdot 5^n \left(\frac{6}{25}\right) = 5^n$$

$$A = \frac{25}{6}$$

putting the value of A in equation (2)

$$a_n = \frac{25}{6} \cdot (5)^n$$

$$a_n^{(p)} = \frac{25}{6}(5)^n = \frac{1}{6} 5^{n+2}$$

total solution -

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$\boxed{a_n = C_1 (2)^n + C_2 (3)^n + \frac{1}{6} \cdot 5^{n+2}}$$

Problem = 2

Solve the recurrence relation $a_n - 7a_n + 10a_{n-2} = 3^n$; given that $a_0 = 0$ and $a_1 = 1$

Solution - Given recurrence relation -

$$a_n - 7a_n + 10a_{n-2} = 3^n \quad \text{---(1)}$$

The characteristic equation is -

$$a_n - 7a_n + 10a_{n-2} = 0$$

$\because a_n = m^n$

$$m^2 - 7m + 10 = 0$$

$m^2 - 5m - 2m + 10 = 0$

$m(m-5) - 2(m-5) = 0$

$(m-5)(m-2) = 0$

$m = 2, 5$

Homogeneous Solution –

$$a_{(H)} = C_1(2)^n + C_2(5)^n$$

Since $f(n) = 3^n$ and 3 is not the root of homogeneous solution. then we will be assume the particular solution.

$$a_{(P)} = A \cdot (3)^n \quad \text{——— (ii)}$$

putting in equation (i)

$$A \cdot (3)^n - 7A \cdot (3)^{n-1} + 10 \cdot A(3)^{n-2} = (3)^n$$

$$A \cdot (3)^n \left(1 - \frac{7}{3} + \frac{10}{9}\right) = 3^n$$

$$A = \frac{9}{9 - 21 + 10} \qquad A = \frac{-9}{2}$$

putting in equation (ii)

$$a_{(P)} = \frac{-9}{2}(3)^n$$

Particular Solution –

$$a_n = C_1(2)^n + C_2(5)^n - \frac{9}{2}(3)^n$$

putting $a_0 = 0$ i.e $a_n = 0$ and $n = 0$

$$\therefore 0 = C_1(2)^0 + C_2(5)^0 - \frac{9}{2} \times (3)^0$$

$$= C_1 + C_2 - \frac{9}{2}$$

$$C_1 + C_2 = \frac{9}{2} \quad \text{——— (iii)}$$

putting $a_1 = 1$ i.e $a_n = 1$ and $n = 1$

$$1 = c_1 \cdot (2)^1 + c_2 (5)^1 - \frac{9}{2} \times 3$$

$$2c_1 + 5c_2 = \frac{29}{2} \quad \text{---(iv)}$$

from equation (iii) and (iv)

$$2c_1 + 5c_2 = \frac{29}{2}$$

$$2c_1 + 2c_2 = 9$$

$$3c_2 = \frac{29}{2} - 9$$

$$3c_2 = \frac{29 - 18}{2} = \frac{11}{2}$$

$$\boxed{c_2 = \frac{11}{6}}$$

putting in equation (iii)

$$\frac{11}{6} + c_1 = \frac{9}{2}$$

$$c_1 = \frac{9}{2} - \frac{11}{6} = \frac{34 - 22}{2 \times 6} = \frac{12}{2 \times 6} = \frac{16}{6} = \frac{8}{3}$$

putting in equation

$$\boxed{a_n = \frac{8}{3}(2)^n + \frac{11}{6}(5)^n - \frac{9}{2}(3)^{n+2}}$$

Case=12 When $f(n) = b^n$ and b is a root of characteristic equation.

1. If multiplicity of b is one i.e b is a root of multiplicity 1, then, Suppose the trial solution is $a_n = A \cdot n \cdot b^n$
2. If multiplicity of b is two i.e b is a root of multiplicity 2 then, Suppose the trial solution is

$$a_n = A n^2 b^n.$$

Problem=13

Solve the recurrence relation

$$a_n - 3a_{n-1} + 2a_{n-2} = (2)^n$$

Ans given recurrence relation—

$$a_n - 3a_{n-1} + 2a_{n-2} = (2)^n \quad \text{— (1)}$$

characteristic equation—

$$m^2 - 3m + 2 = 0$$
$$m^2 - 2m - m + 2 = 0$$
$$m(m-2) - 1(m-2) = 0$$
$$(m-2)(m-1) = 0$$
$$m = 1, 2$$

Homogeneous solution—

$$a_n^{(H)} = C_1 (1)^n + C_2 (2)^n \quad \text{— (11)}$$

Since $f(n) = 2^n$ and 2 is a root of characteristic equation with one multiplicity. So we will assume the particular solution —

$$a_n^{(P)} = A \cdot n 2^n \quad \text{— (2)}$$

putting in equation (1)

$$An2^n - 3A(n-1)2^{n-1} + 2A(n-2)2^{n-2} = 2^n$$

$$A2^n \left(n - \frac{3(n-1)}{2} + \frac{2(n-2)}{2} \right) = 2^n$$

$$A \left(\frac{2n - 3n + 3 + 2n - 4}{2} \right) = 1$$

$A n \cdot 2^n - \frac{3}{2} A n 2^n + \frac{3A}{2} 2^n + \frac{1}{2} A n \cdot 2^n - A 2^n = 2^n$

$\frac{1}{2} A 2^n = 2^n$

$\boxed{A = 2}$

putting in equation (iii)

$a_n^{(p)} = 2n\,2^n$

particular solution-

$\boxed{a_n = c_1 (1)^n + c_2 (2)^n + 2n 2^n}$

or

$\boxed{a_n = c_1 + c_2 (2)^n + n \cdot 2^{n+1}}$

Problem = 4

b) Solve the recurrence relation $a_n - 4a_{n-1} + 4a_{n-2} = 2^n$

given recurrence relation-

$a_n - 4a_{n-1} + 4a_{n-2} = 2^n$ — (i)

characteristic equation-

$m^2 - 4m + 4 = 0$

$m^2 - 2m - 2m + 4 = 0$

$m(m-2) - 2(m-2) = 0$

$(m-2)(m-2) = 0$

$m = 2, 2$

Homogeneous solution-

$\boxed{a_n^{(H)} = (c_1 + c_2 n) 2^n}$ — (ii)

Since $f(n) = 2^n$ and 2 is a root of characteristic equation with 2 multiplicity. Then we will write particular solution

$a_n^{(P)} = A n^2 (2)^n$ — (iii)

putting in equation (ii) —

$An^2 (2)^n - 4A(n-1)^2 (2)^{n-1} + 4A(n-2)^2 2^{n-2} = 2^n$

$A n^2 (2)^n - 2 \cdot A (n-1)^2 (2)^n + A (n-2)^2 2^n = 2^n$

$2^n (A n^2 - 2A (n-1)^2 + A (n-2)^2) = 2^n$

$A \left[n^2 - 2 (n^2 + 1 - 2n) + (n^2 + 4 - 4n) \right] = 1$

$A \left[n^2 - 2n^2 + 2 + 4n + n^2 + 4 - 4n \right] = 1$

$A \times 2 = 1$

$A = \frac{1}{2}$

putting in equation (iii) -

$a_n(p) = \frac{1}{2} n^2 (2)^n$

Particular solution -

$a_n^{(p)} = n^2 \cdot (2)^{n-1}$

total solution -

$$a_n = (C_1 + C_2 n)(2)^n + n^2 \cdot 2^{n-1}$$

Solve the recurrence relation $a_n - 4a_{n-1} + 3a_{n-2} = 5^n$

Ans given recurrence relation -

$a_n - 4a_{n-1} + 3a_{n-2} = 5^n$

characteristic equation -

$m^2 - 4m + 3 = 0$

$m^2 - 3m - m + 3 = 0$

$m(m-3) - 1(m-3) = 0$

$(m-3)(m-1) = 0$

$m = 1, 3$

Since $f(n) = g^n$ and 5 is not match to root of characteristic equation. Then we will write particular -

$a_n(p) = A \cdot (5)^n$

putting in equation (i)

$$A5^n - 4A5^{n-1} + 3A5^{n-2} = 5^n$$

$$A5^n - \frac{4}{5}A5^n + \frac{3A}{25}5^n = 5^n$$

$$A5^n\left(1 - \frac{4}{5} + \frac{3}{25}\right) = 5^n$$

$$A\left(\frac{25 - 20 + 3}{25}\right) = 1$$

$$A = \frac{25}{8}$$

Particular solution -

$$a_n^{(p)} = A(5)^n$$

$$= \frac{25}{8}(5)^n$$

$$= \frac{5^{n+2}}{8}$$

total solution -

$$a_n = c_1(1)^n + c_2(3)^n + \frac{5^{n+2}}{8}$$

//

case=3 When $f(n)$ is a polynomial in n

[i] If $f(n)$ is first degree polynomial in n then trial solution is $a_n^{(p)} = A_0 + A_1 n$

[ii] If $f(n)$ is second degree polynomial in n then trial solution

$$a_n^{(p)} = A_0 + A_1 n + A_2 n^2$$

Problem=1

Solve the recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 2 + n$ given that $a_0 = 1$ and $a_1 = 1$

Ans Given recurrence relation -

$$a_n - 5a_{n-1} + 6a_{n-2} = 2 + n \quad \text{---(i)}$$

characteristic equation -

$$m^2 - 5m + 6 = 0$$

$m^2 - 3m - 2m + 6 = 0$

$m(m-3) - 2(m-3) = 0$

$(m-3)(m-2) = 0$

$m = 2, 3$

Homogeneous solution-

$$a_n^{(h)} = c_1 (2)^n + c_2 (3)^n \quad \text{— (ii)}$$

Since $f(n) = 2 + n$ is a polynomial equation of degree one, then we will assume the particular solution-

$$a_n^{(p)} = A_0 + A_1 n \quad \text{— (iii)}$$

putting in equation (i)

$$[A_0 + A_1 n] - 5[A_0 + A_1 (n+1)] + 6[A_0 + A_1 (n-2)] = 2 + n$$

$$A_0 + A_1 n - 5A_0 - 5A_1 n - 5A_1 + 6A_0 + 6A_1 n - 12A_1 = 2 + n$$

$$2A_0 + 2A_1 n - 7A_1 = 2 + n$$

$$(2A_0 - 7A_1) + 2A_1 n = 2 + n$$

comparision of both side-

$$2A_0 - 7A_1 = 2 \quad -$$

$$2A_1 = n$$

$$A_1 = \frac{1}{2}$$

$$2A_0 - 7 \times \frac{1}{2} = 2$$

$$2A_0 = 2 + \frac{7}{2}$$

$$A_0 = \frac{11}{2} \times \frac{1}{2}$$

$$A_0 = \frac{11}{4}$$

putting in equation (iii)

Particular solution-

$$a_n^{(p)} = \frac{11}{4} + \frac{1}{2} n$$

total solution-

$$a_n = c_1 (2)^n + c_2 (3)^n + \frac{11}{4} + \frac{1}{2} n \quad \text{— (iv)}$$

putting $a_0 = 1$ when $a_n = 4$ when $n = 0$

$$1 = C_1 + C_2 + \frac{11}{4}$$

$$C_1 + C_2 = 1 - \frac{11}{4}$$

$$C_1 + C_2 = \frac{-7}{4} \quad \text{(V)}$$

putting $a_1 = 1$ when $a_n = 1$, when $n = 1$

$$1 = 2C_1 + 3C_2 + \frac{11}{4} + \frac{1}{2}$$

$$2C_1 + 3C_2 = 1 - \frac{11}{4} - \frac{1}{2}$$

$$= \frac{4 - 11 - 2}{4} = \frac{4 - 13}{4} = \frac{-9}{4}$$

$$2C_1 + 3C_2 = \frac{-9}{4} \quad \text{(VI)}$$

from equation (V) and (VI)

$$2C_1 + 3C_2 = \frac{-9}{4}$$

$$2C_1 + 2C_2 = -\frac{7}{2}$$

$$\frac{7}{2} - \frac{9}{4}$$

$$\frac{14 - 9}{4} = \frac{5}{4}$$

$$C_2 = \frac{5}{4}$$

putting in equation (V)

$$C_1 + \frac{5}{4} = \frac{-7}{4}$$

$$C_1 = \frac{-5}{4} - \frac{7}{4} = \frac{-12}{4} = -3$$

$$C_1 = -3$$

putting the value of C_1 and C_2 in equation (IV)

$$a_n = -3(2)^n + \frac{5}{4}(3)^n + \frac{11}{4} + \frac{1}{2}n$$

Problem - solve the recurrence relation $a_n - 4a_{n-1} + 4a_{n-2} = (n+1)^2$

Sol given recurrence relation -

$$a_n - 4a_{n-1} + 4a_{n-2} = (n+1)^2 \quad \text{---(i)}$$

The characteristic equation is -

$$a_n - 4a_{n-1} + 4a_{n-2} = 0$$

$$m^2 - 4m + 4 = 0$$
$$m^2 - 2m - 2m + 4 = 0$$
$$m(m-2) - 2(m-2) = 0$$
$$(m-2)(m-2) = 0$$
$$m = 2, 2$$

Homogeneous equation -

$$a_n^{(H)} = (C_1 + C_2 n)(2)^n$$

Since $f(n) = (n+1)^2$ is a polynomial equation of two degree, then we will assume the particular solution of equation -

$$a_n^{(P)} = 2n + 1 + n^2$$
$$a_n^{(P)} = d_1 n^2 + d_2 n + d_3 \quad \text{---(ii)}$$

putting in equation (i)

$$(d_1 n^2 + d_2 n + d_3) - 4\left[d_1(n-1)^2 + d_2(n-1) + d_3\right] + 4\left[d_1(n-2)^2 + d_2(n-2) + d_3\right] = n^2 + 2n + 1$$

$$d_1 n^2 + d_2 n + d_3 - 4\left[d_1(n^2 + 1 - 2n) + d_2(n-1) + d_3\right] + 4\left[d_1(n^2 + 4 - 4n) + d_2 n - 2d_2 + d_3\right] = n^2 + 2n + 1$$

$$d_1 n^2 + d_2 n + d_3 - 4d_1 n^2 + 4d_1 + 8d_1 n - 4d_2 n + 4d_2 + d_3 + 4d_1 n^2 + 16d_1 - 16d_1 n + d_2 n + 2d_2 + d_3 = n^2 + 2n + 1$$

$$d_1 n^2 - 8d_1 n - 2d_2 n + 12d_1 + 5d_2 + 3d_3 = n^2 + 2n + 1$$

$$\boxed{d_1 = 1}$$

$-(3d_1 + 2d_2) = 2$

$-(3\times 1 + 2d_2) = 2$

$3 + 2d_2 = -2$

$2d_2 = -10$

$\boxed{d_2 = -5}$

$12d_1 + 5d_2 + 3d_3 = 1$

$12\times 1 + 5\times -5 + 3d_3 = 1$

$12 - 25 + 3d_3 = 1$

$3d_3 = 25 + 1 - 12$

$3d_3 = 26 - 12$

$3d_3 = 14$

$d_3 = \frac{14}{3}$

putting in equation (ii)

$a_n(p) = d_1 n^2 + (-5)n + \frac{14}{3}$

$a_n(p) = n^2 - 5n + \frac{14}{3}$

total solution -

$$\boxed{a_n = (c_1 + c_2 n)(2)^n + n^2 - 5n + \frac{14}{3}}$$

Problem -3

Solve the recurrence relation $a_n + 6a_{n-1} + 9a_{n-2} = 3$ given that $a_0 = 0$ and $a_1 = 1$

Ans

given recurrence relation

$$a_n + 6a_{n-1} + 9a_{n-2} = 3$$

characteristic relation -

$$m^2 + 6m + 9 = 0 \quad \text{---(i)}$$

$m^2+3m+3m+9=0$

$m(m+3)+3(m+3)=0$

$(m+3)(m+3)=0$

$m=-3,-3$

Homogeneous Solution -

$$a_n^{(h)} = (c_1+c_2 n)(-3)^n \quad \text{— (ii)}$$

Since $f(n)=3$ is a polynomial equation of zero degree then particular equation -

$$a_n^{(p)} = d_1$$

putting in equation (i)

$$d_1 + 6d_1 + 9d_1 = 3$$

$$16d_1 = 3$$

$$d_1 = \frac{3}{16}$$

Particular Solution

$$a_n^{(p)} = \frac{3}{16}$$

total solution of given recurrence relation -

$$\boxed{a_n = (c_1+c_2 n)(-3)^n + \frac{3}{16}} \quad \text{— (iii)}$$

putting $a_0=0$ and $n=0$ in equation (iii)

$$0 = (c_1+c_2\times 0)(-3)^0 + \frac{3}{16}$$

$$= c_1 + \frac{3}{16}$$

$$\boxed{c_1 = \frac{-3}{16}}$$

putting $a_1 = 1$ where $n = 1$

$$1 = (C_1 + C_2)(-3)^1 + \frac{3}{16}$$

$$1 = -3C_1 - 3C_2 + \frac{3}{16}$$

$$3C_2 = -3 \times \frac{-3}{16} + \frac{3}{16} - 1$$

$$3C_2 = \frac{9}{16} + \frac{3}{16} - 1$$

$$3C_2 = \frac{12 - 16}{16} = \frac{-4}{16} = -\frac{1}{4}$$

$$C_2 = -\frac{1}{12}$$

putting in equation (iii)

$$a_n = \left(\frac{-3}{16} - \frac{1}{12}n\right)(-3)^n + \frac{3}{16}$$

$$a_n = -\left(\frac{3}{16} + \frac{n}{12}\right)(-3)^n + \frac{3}{16}$$

Solve the following recurrence relation:-

[i] $a_n - 5a_{n-1} + 6a_{n-2} = n(n-1)$

Ans given recurrence relation -

$$a_n - 5a_{n-1} + 6a_{n-2} = n^2 - n \quad \text{— (i)}$$

characteristic equation -

$$m^2 - 5m + 6 = 0$$

$$m^2 - 3m - 2m + 6 = 0$$

$$m(m-3) - 2(m-3) = 0$$

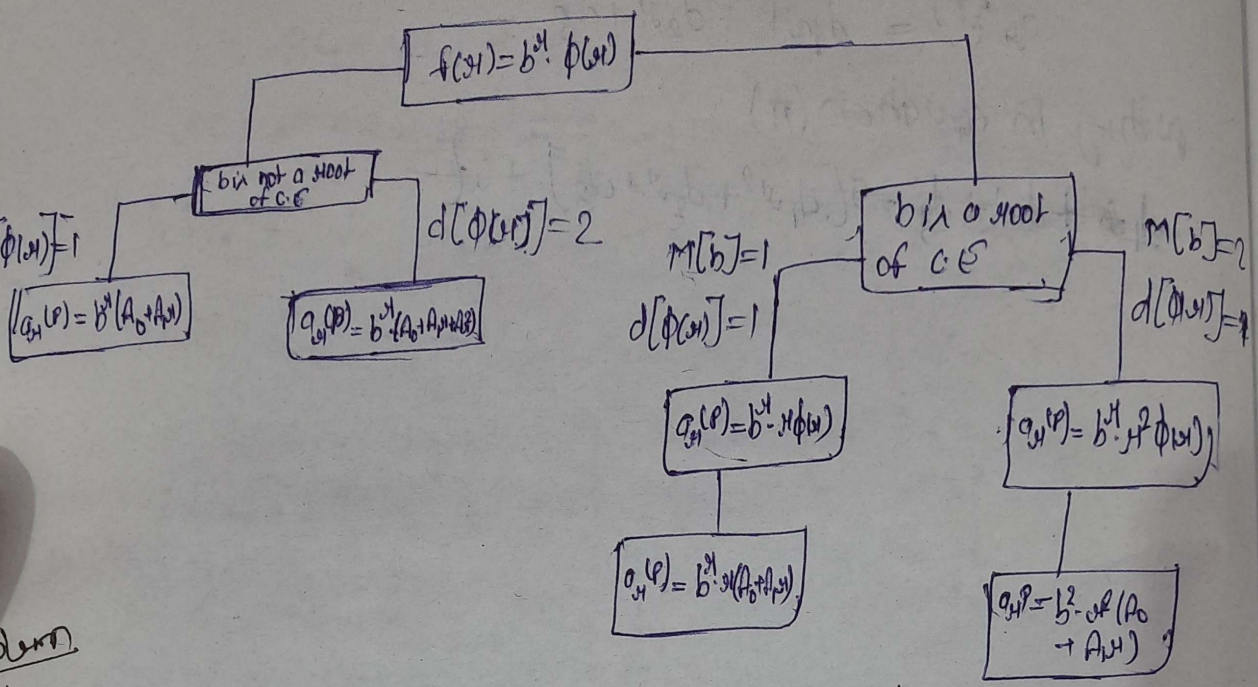
$$(m-3)(m-2) = 0$$

$$m = 2, 3$$

Homogeneous solution -

$$a_n(H) = C_1(2)^n + C_2(3)^n \quad \text{— (ii)}$$

1. **Case-II** When $f(n) = b^n \cdot \phi(n)$ where $\phi(n)$ is a polynomial in n and b is a constant.



Problem

Solve the recurrence relation $a_n + a_{n-1} = 3n \cdot 2^n$

Ans given recurrence relation-

$$a_n + a_{n-1} = 3n \cdot 2^n \quad \text{---(i)}$$

characteristic equation-

$$m + 1 = 0$$
$$m = -1$$

Homogeneous solution.

$$a_n^{(h)} = c_1 (-1)^n \quad \text{---(ii)}$$

Since $f(n) = 3n \cdot 2^n$ is polynomial equation And 2 is not root of characteristic equation. So particular equation is-

$$a_n^{(p)} = 2^n (A_1 + A_2 n) \quad \text{---(iii)}$$

putting in equation (i)

$$2^n (A_1 + A_2 n) + \frac{2^{n-1}[A_1 + A_2 (n-1)]}{2} = 3n \cdot 2^n$$

$$2^n(A_1+A_2n)+\frac{2^n}{2}(A_1+A_2n-A_2)=3n\cdot 2^n$$

$$2^n\left[A_1+A_2n+\frac{A_1}{2}+\frac{A_2n}{2}-\frac{A_2}{2}\right]=3n\cdot 2^n$$

$$\frac{3A_1}{2}+\frac{3A_2n}{2}-\frac{A_2}{2}=3n$$

$$\frac{3A_2n}{2}+\frac{3A_1}{2}-\frac{A_2}{2}=3n$$

Taking comparision on both side -

$$\frac{3A_2}{2}=3 \qquad \boxed{A_2=2}$$

$$\frac{3}{2}A_1=\frac{A_2}{2} \qquad \boxed{A_1=\frac{A_2}{3}=\frac{2}{3}}$$

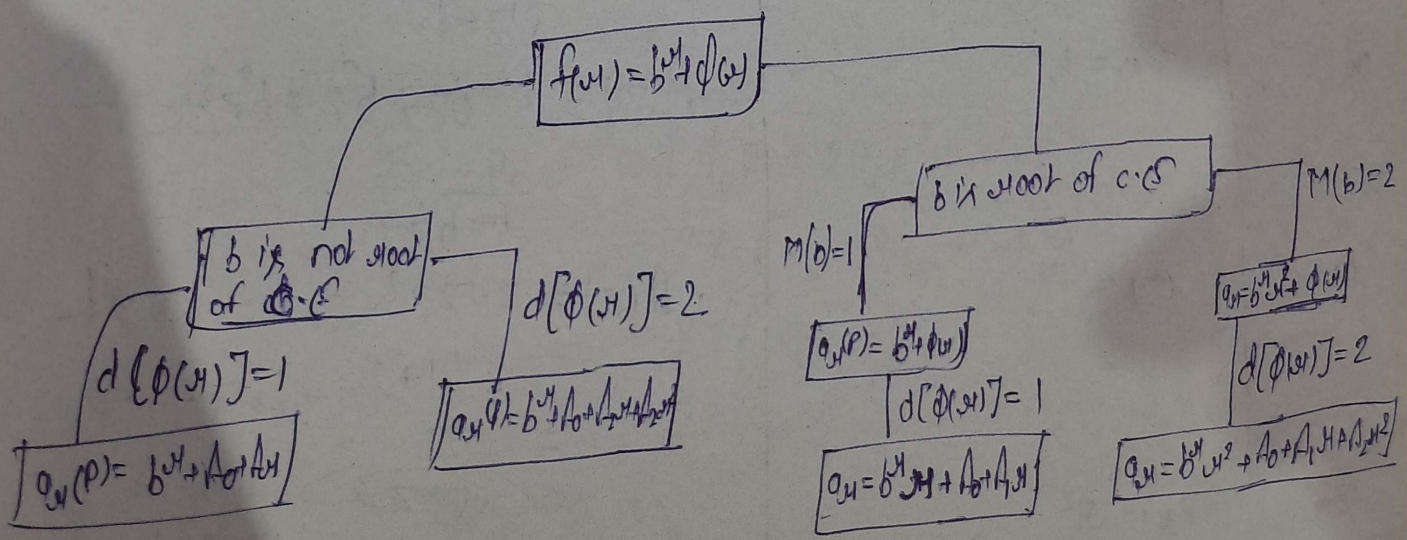
putting in equation (iii)

$$a_n(P)=2^n\left(2+\frac{2}{3}n\right)$$

total solution of given recurrence relation -

$$\boxed{a_n=C_1(-1)^n+2^n\left(2+\frac{2n}{3}\right)}$$

Case 5 When $f(n)=b^n\phi(n)$, where $\phi(n)$ is a polynomial in n and b is constant.



* Generating function *

If $a = (a_0, a_1, a_2, a_3 \ldots a_n \ldots)$ is a discrete numeric function, then an infinite series in terms of a parameter z written as

$$A(z) = a_0 + a_1 z^1 + a_2 z^2 + a_3 z^3 + a_4 z^4 + \ldots$$

is said to be a generating function of the numeric function a and it is also written in the form.

$$A(z) = \sum_{n=0}^{\infty} a_n z^n$$

some important results

S.NO	Numeric function (General Term of numerical function)	Generating function $A(z)$
1	$a_n = 1$	$A(z) = 1 + z + z^2 + \ldots + z^n + \ldots$ $= (1-z)^{-1}$ $= \frac{1}{(1-z)}$
2.	$a_n = k$	$A(z) = k + kz + kz^2 + \ldots + kz^n + \ldots$ $= k(1 + z + z^2 + \ldots)$ $= \frac{k}{(1-z)}$
3.	$a_n = b^n$	$A(z) = 1 + bz + b^2z^2 + b^3z^3 + \ldots$ $= \frac{1}{1-bz}$
4.	$a_n = n$	$A(z) = z + 2z^2 + 3z^3 + \ldots + nz^n + \ldots$ $= z(1 + 2z + 3z^2 + \ldots)$ $= \frac{z}{(1-z)^2}$
5.	$a_n = n^2$	$A(z) = z + 2^2z^2 + 3^2z^3 + \ldots$ $= z(1 + 2^2z + 3^2z^2 + \ldots)$ $= \frac{z(1+z)}{(1-z)^3}$

6.	$a_n = n(n+1)$	$A(z) = 1\cdot 2z + 2\cdot 3z^2 + 3\cdot 4z^3 + \cdots$ $= \frac{2z}{(1-z)^3}$
7.	$a_n = k \cdot b^n$	$A(z) = k + kbz + k\cdot b^2 z^3 + k\cdot b^3 z^4 + \cdots$ $= k(1 + bz + b^2 z^3 + b^3 z^4 + \cdots)$ $= \frac{k}{1-bz}$ //

Q:1 Find the generating function for that sequence of relation $a_n + 2a_{n-1} = 0$ with an initial condition $a_0 = 5$

Ans given recurrence relation -

$$a_n + 2a_{n-1} = 0$$

$$\sum_{n=1}^{\infty} (a_n z^n + 2a_{n-1} z^n) = 0 \quad \text{---(1)}$$

Standard generating sequence -

$$G(A,z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$$

first term -

$$\sum_{n=1}^{\infty} a_n z^n = a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 + \cdots$$

$$\sum_{n=1}^{\infty} a_n z^n = G(A,z) - a_0$$

second term -

$$\sum_{n=1}^{\infty} 2a_{n-1} z^n = 2a_0 z + 2a_1 z^2 + 2a_2 z^3 + \cdots$$

$$= 2z(a_0 + a_1 z + a_2 z^2 + \cdots)$$

$$= 2z(G(A,z))$$

$$= 2z\,G(A,z)$$

putting in equation (1)

$$G(A,z) - a_0 + 2z\,G(A,z) = 0$$

$$G(A,z)(1 + 2z) = 5$$

$$G(A,z) = \frac{5}{(1+2z)}$$

$$G(A,z) = \frac{5}{(1-(-2)z)}$$

$$= 5(-2)^n$$

$$\boxed{G(A,z) = 5(-2)^n}$$

Solve the recurrence relation $a_n - 6a_{n-1} + 8a_{n-2} = 0$ with the initial condition $a_0 = 5$, $a_1 = 8$

Ans given recurrence relation

$$a_n - 6a_{n-1} + 8a_{n-2} = 0$$

generating equation –

$$\sum_{n=2}^{\infty} (a_n - 6a_{n-1} + 8a_{n-2}) z^n = 0$$

$$\sum_{n=2}^{\infty} a_n z^n - 6\sum_{n=2}^{\infty} a_{n-1} z^n + 8\sum_{n=2}^{\infty} a_{n-2} z^n = 0 \quad \text{---(1)}$$

Standard generated equation

$$G(A,z) = \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \cdots$$

first term –

$$\sum_{n=2}^{\infty} a_n z^n = a_2 z^2 + a_3 z^3 + a_4 z^4 + \cdots$$

$$= G(A,z) - a_0 + a_1 z$$

second term –

$$\sum_{n=2}^{\infty} a_{n-1} z^n = a_1 z^2 + a_2 z^3 + a_3 z^4 + a_4 z^5 + \cdots$$

$$= z(a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 + \cdots)$$

$$= z[G(A,z) - a_0]$$

Third term –

$$\sum_{n=2}^{\infty} a_{n-2} z^n = a_0 z^2 + a_1 z^3 + a_2 z^4 + \cdots$$

$$= z^2 (a_0 + a_1 z + a_2 z^2 + \cdots)$$

$$= z^2 G(A,z)$$

putting in equation (i)

$$G(A,z) - a_0 - a_1 z - 6z[G(A,z) - a_0] + 8z^2 G(A,z) = 0$$

$$G(A,z) - 5 - 8z - 6zG(A,z) + 30z + 8z^2 G(A,z) = 0$$

$$G(A,z)(1 - 6z + 8z^2) + 22z - 5 = 0$$

$$G(A,z) = \frac{5 - 22z}{8z^2 - 6z + 1} = \frac{5 - 22z}{8z^2 - 4z - 2z + 1} = \frac{5 - 22z}{4z(2z-1) - 1(2z-1)}$$

$$G(A,z) = \frac{5 - 22z}{(4z-1)(2z-1)} = \frac{22z - 5}{(1-4z)(1-2z)}$$

$$\frac{22z - 5}{(1-4z)(1-2z)} = \frac{A}{(1-4z)} + \frac{B}{(1-2z)}$$

$$22z - 5 = A(1-2z) + B(1-4z) \quad \text{— (ii)}$$

if $1 - 2z = 0$

$z = \frac{1}{2}$

putting in equation -

$$22 \times \frac{1}{2} - 5 = A\left(1 - 2 \times \frac{1}{2}\right) + B\left(1 - 4 \times \frac{1}{2}\right)$$

$$11 - 5 = B(1-2)$$

$$-B = 6$$

$$\boxed{B = -6}$$

If $1 - 4z = 0$

$z = \frac{1}{4}$

$$22 \times \frac{1}{4} - 5 = A\left(1 - 2 \times \frac{1}{4}\right) + B\left(1 - 4 \times \frac{1}{4}\right)$$

$$\frac{11}{2} - 5 = A \times \frac{1}{2}$$

$$\frac{11 - 10}{2} = \frac{A}{2}$$

$$\boxed{A = 1}$$

$$G(A,z) = \frac{1}{(1-4z)} + \frac{(-6)}{(1-2z)}$$

$$= 4^n - 6(2)^n$$

$$\boxed{G(A,z) = 4^n - 6(2)^n}$$

Problem 3 Solve the following recurrence relation using generating function $a_n - 2a_{n-1} - 3a_{n-2} = 0$ initial condition $a_0 = 3, a_1 = 1, n \geq 2$

Ans given recurrence relation –

$$a_n - 2a_{n-1} - 3a_{n-2} = 0 \rightarrow$$

generating form –

$$\sum_{n=2}^{\infty} (a_n - 2a_{n-1} - 3a_{n-2}) z^n = 0$$

$$\sum_{n=2}^{\infty} a_n z^n - 2\sum_{n=2}^{\infty} 2a_{n-1} - 3\sum_{n=2}^{\infty} a_{n-2} z^n = 0 \quad \text{—— (1)}$$

standard generating equation

$$G(A,z) = \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \text{———}$$

first term –

$$\sum_{n=2}^{\infty} a_n z^n = a_2 z^2 + a_3 z^3 + a_4 z^4 + \text{———}$$

$$= G(A,z) - a_0 - a_1 z$$

second term

$$\sum_{n=2}^{\infty} a_{n-1} z^n = a_1 z^2 + a_2 z^3 + a_3 z^4 + \text{———}$$

$$= z(a_1 z + a_2 z^2 + a_3 z^3 + \text{———})$$

$$= z(G(A,z) - a_0)$$

third term

$$\sum_{n=2}^{\infty} a_{n-2} z^n = a_0 z^2 + a_1 z^3 + a_2 z^4 + \text{———}$$

$$= z^2(a_0 + a_1 z + a_2 z^2 + \text{———})$$

$$= z^2 G(A,z)$$

putting in equation (1)

$$G(A,z) - a_0 - a_1 z - 2z[G(A,z) - a_0] - 3z^2 G(A,z) = 0$$

$$G(A,z) - g_0 - g_1 z - 2z G(A,z) + 2zg_0 - 8z^2 G(A,z) = 0$$

$$G(A,z)(1 - 2z - 8z^2) - 3 - z + 2z \times 3 = 0$$

$$G(A,z)(-8z^2 - 2z + 1) = z - 6z + 3$$

$$G(A,z) = \frac{3 - 5z}{(-8z^2 - 2z + 1)}$$

$$G(A,z) = \frac{3 - 5z}{8z^2 - 4z + 2z + 1} = \frac{3 - 5z}{4z(2z - 1) + 1(2z}$$

$$G(A,z) = \frac{5z - 3}{8z^2 + 2z - 1} = \frac{5z - 3}{8z^2 + 4z - 2z - 1} = \frac{5z - 3}{4z(2z+1) - 1(2z+1)}$$

$$G(A,z) = \frac{5z - 3}{(2z+1)(4z-1)}$$

$$\frac{5z-3}{(2z+1)(4z-1)} = \frac{A}{(2z+1)} + \frac{B}{(4z-1)}$$

$$5z - 3 = A(4z-1) + B(2z+1)$$

$$5z - 3 = A(4z-1) + B(2z+1)$$

if $4z - 1 = 0$

$4z = 1$

$z = \frac{1}{4}$

$$5 \times \frac{1}{4} - 3 = A\left(1 \times \frac{1}{4} - 1\right) + B\left(2 \times \frac{1}{4} + 1\right)$$

$$\frac{5}{4} - 3 = B \times \frac{3}{2}$$

$$\frac{-7}{4_2} = B \times \frac{3}{2}$$

$$\boxed{B = -\frac{7}{6}}$$

if $2z + 1 = 0$

$2z = -1$

$z = -\frac{1}{2}$

$$5x - \frac{1}{2} - 3 = A\left(4x - \frac{1}{2} - 1\right) + B\left(2x - \frac{1}{2} + 1\right)$$

$$-\frac{5}{2} - 3 = -3A$$

$$\frac{-11}{2} = -3A$$

$$\boxed{A = \frac{11}{6}}$$

$$G(A,z) = \frac{\frac{11}{6}}{[1 - (-2)z]} + \frac{\frac{7}{6}}{(1 - 4z)}$$

$$\boxed{G(A,z) = \frac{11}{6}(-2)^n + \frac{7}{6}(4)^n}$$

Q=2 Solve the recurrence relation using generating function $a_n - a_{n-1} - 2a_{n-2} = 0$, if intial condition $a_0 = 0, a_1 = 1$

Ans generating function -

$$a_n - a_{n-1} - 2a_{n-2} = 0$$

generating form -

$$\sum_{n=2}^{\infty} (a_n - a_{n-1} - 2a_{n-2}) z^n = 0$$

$$\sum_{n=2}^{\infty} a_n z^n - \sum_{n=2}^{\infty} a_{n-1} z^n - 2\sum_{n=2}^{\infty} a_{n-2} z^n = 0 \quad \text{— (1)}$$

generating equation -

$$G(A,z) = \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z^1 + a_2 z^2 + a_3 z^3 + \cdots$$

first term -

$$\sum_{n=2}^{\infty} a_n z^n = a_2 z^2 + a_3 z^3 + a_4 z^4 + \cdots$$

$$= G(A,z) - a_0 - a_1 z$$

second term

$$\sum_{n=2}^{\infty} a_{n-1} z^n = a_1 z^2 + a_2 z^3 + a_3 z^4 + \cdots$$

$$= z(a_1 z + a_2 z^2 + a_3 z^3 + \cdots)$$

$$= z \times [G(A,z) - a_0]$$

Third term

$$\sum_{n=2}^{\infty} a_{n-2} z^n = a_0 z^2 + a_1 z^3 + a_2 z^4 + \cdots$$

$$= z^2 (a_0 + a_1 z + a_2 z^2 + \cdots)$$

$$= z^2 G(A,z)$$

putting in equation (i)

$$G(A,z) - a_0 - a_1 z - z G(A,z) + z a_0 - 2z^2 G(A,z) = 0$$

$$G(A,z) - 0 - z - z G(A,z) + 0 - 2z^2 G(A,z) = 0$$

$$G(A,z)(1 - z - 2z^2) = z$$

$$G(A,z) = \frac{z}{(1-z-2z^2)} = \frac{z}{(1-2z+z-2z^2)} = \frac{z}{1(1-2z)+z(1-2z)}$$

$$= \frac{z}{(1-2z)(1+z)}$$

$$\frac{z}{(1-2z)(1+z)} = \frac{A}{(1-2z)} + \frac{B}{(1+z)}$$

$$z = A(1+z) + B(1-2z)$$

if $1+z=0$

$z=-1$

$\therefore -1 = A(1-1) + B(1+2)$

$3B = -1$

$\boxed{B = -\frac{1}{3}}$

if $1-2z=0$

$1=2z$

$z=\frac{1}{2}$

$\frac{1}{2} = A\left(1+\frac{1}{2}\right)$

$\frac{1}{2} = \frac{3}{2}A$

$\boxed{A = \frac{1}{3}}$

$$G(A,z) = \frac{\frac{1}{3}}{(1-2z)} + \frac{-\frac{1}{3}}{(1+z)}$$

$$= \frac{\frac{1}{3}}{(1-2z)} + \frac{-\frac{1}{3}}{[1-(-1)z]}$$

$$\therefore G(A,z) = \frac{1}{3}(2)^n - \frac{1}{3}(-1)^n$$

Permutation & combination

Siddhanta shiromani → Bhaskaracharya-II → Leelavati (daughter)

Aryabhatiya, Mahabhaskariya

Permutation :-

Introduction of counting principle

Fundamental Principle of counting

- Fundamental Multiplication Principle of counting
- Fundamental addition principle of counting

Fundamental Multiplication Principle of counting :- If there are 'm' different ways to doing one event and 'n' different ways of doing another event, then the simultaneous occurance of these events will be given by 'm×n' different ways.

$$\text{total way} = m \times n$$

Fundamental addition principle of counting :- If there are 'm' different ways of doing an event and 'n' other different way to doing same event then the number of way in which the event can be done is given by 'm+n' different ways.

$$\text{total way} = m + n$$

Problem-1

wi

disperse

accomm

Total

total n

n

n

Problem

mailed

Problem-1 Four visitors A, B, C and D arrive at a Town which has 5 hotels. In how many ways can they disperse themselves among 5 hotels if 4 hotels are used to accommodate them.

Ans Total number of visitors = 4 visitors
Total number of hotel = 5 hotel

Total number of possibility to choose total Hotel by A visitors = 5

total number of possibility to choose Hotel by B visitors after choosing A = 4
" " " " " " " C " " " A, B = 3
" " " " " " " " " " " A, B, C = 2

total possibility to choose Hotel by visitors = $5 \times 4 \times 3 \times 2$
$= 120$ //

Problem-2. In how many ways can 5 letters be mailed if there are 3 mailboxes available if each letters can be mailed in any mailbox.

Ans. Total number of letters = 5
Total number of mailbox = 3

~~Letter first~~
The possibility to choose mailbox for send letters first = 3
" " " " " " " " second = 3
" " " " " " " " Third = 3
" " " " " " " " four = 3
" " " " " " " " five = 3

total possibility to choose mailbox = $3 \times 3 \times 3 \times 3 \times 3$
$= 243$

Problem-3 Every telephone number consists of 7 digits. How many telephone number are there which do not include any other digit but 2, 3, 5 & 7?

Ans Total digit which use to construct seven digit number = 4

Total digit which required to consist telephone no = 7

1 digit of telephone number is construct by 4 ways.

Similarly —

Total telephone number which construct = $4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4$

$= 4^7$

Permutation:- Permutation is the collection arrangement of the object in order.

or

The permutation is the number of different arrangement that can be made by picking a number of object from the available n object.

$$^nP_n = \frac{n!}{(n-n)!}$$

Pigeon hole Principle

If (N+1) or more objects are placed into N boxes, then there is atleast one box containing two or more objects.

Ex if 6 colours are used to paint 37 horse. show that at least 7 horse of them will be of same colour.

A $\frac{37}{6} = 6.166$ (Take a highest integer)

$= 7$ //

hence, 7 horse may have a same colour.

Generalized pigeon hole principle :- if 'n' pigeon hole are occupied by kn+1 or more pigeons then atleast one pigeonhole is occupied by k+1 or more pigeon.

Problem ⇒ Find the minimum number of teachers in a college to be sure that four of them are born in same month.

Ans: Let k is the number of teacher who born in same month.

$k+1 = 4$

$k = 3$

Probability of the month $(n) = 12$

$kn+1 = 3 \times 12 + 1$

$= 36 + 1$

$= 37$ //

Problem ⇒ 2 A box contain 10 blue balls, 20 red balls, 8 green balls, 15 yellow balls and 25 white balls. How many balls might be choose to ensure that we have 12 balls of the same colour.

Sol Let k is the number of balls of same colour.

$k+1 = 12$

$k = 11$

Total number of balls = 5

$kn+1 = 11\times 5+1$

$= 55+1$

$= 56$ //

Problem-3. Prove that among 100000 people there are two who are born on same time.

Sol Let $P = \{P_1, P_2, \cdots P_{100000}\}$ be the set of people.

Let $H = \{H_1, H_2, \cdots H_{24}\}$ be the set of all hours in day

By extended pigeon hole principle, there are atleast

$\left[\frac{|P|}{|H|}\right]$ person during the same hour.

$= \frac{100000}{24} = 4167$

Now $Q = \{Q_1, Q_2, Q_3 \cdots Q_{4167}\}$ and $M = \{M_1, M_2 \cdots M_{60}\}$

The number of person born in same minute are atleast

$= \left(\frac{4167}{60}\right) = 70$

Again $R = \{R_1, R_2, R_3\}$ & $S = \{S_1, S_2, \cdots S_{60}\}$

Then no. of person born in same second are atleast

$= \left[\frac{70}{60}\right] = 2$ //

Selection or combination

Selection of Different objects

Ex $A_1, A_2, A_3, A_4 \ldots$

Selection of alike objects

Ex $A, A, A, A \ldots$

combination/Selection of different objects

Q=1 consider four different balls A, B, C, D

(i) In how many ways we can select one ball.

(ii) " " " " " " " two balls.

(iii) " " " " " " " three balls!

Ans (i) The total ways to select one ball from the four balls = 4 ways

(ii) " " " " " two " " " " = 6 ways

(iii) " " " " " three " " " " ——

ABC, BCD, ABD, ACD

total ways = 4 ways

In General

'n' different balls select 'r' different balls total number of ways $= \frac{n!}{r!(n-r)!}$

$$^nC_r = \frac{n!}{r!(n-r)!}$$

where

^nC_r = Total number of ways of selection of 'r' different object out of n- different objects.

Properties:-

1. $^nC_r \times r! = {}^nP_r$ — arrangement of 'r' objects; selection of 'r' diff object out of n-diff objects; arrangement of n-diff objects taken 'r' at a time.

$^nC_r = {}^nC_{n-r}$

Prove

$$^nC_r = \frac{n!}{r!(n-r)!}$$

$$r! \times {}^nC_r = \frac{n!}{(n-r)!}$$

where $^nP_r = \frac{n!}{(n-r)!}$

$^nP_r = r! \times {}^nC_r$

Problem-2 If

To fill

can be giv

Ans

To

The to

The

Problem ⇒ There are m points on a straight line AB & n points on the line AC none of them being the point A. The Triangles are formed with these points as vertices, when.

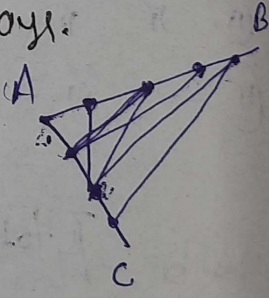
(i) 'A' is excluded. Find the total number of ways.

(ii) 'A' is included. Find the total number of ways.

Ans (1) 2 point on line AB & 1 point on line AC = $^mC_2 \times {}^nC_1$

1 point on line AB & 2 point on line AB = $^mC_1 \times {}^nC_2$

total no. of Triangle = $^mC_2 \times {}^nC_1 + {}^mC_1 \times {}^nC_2$



(2).

1 point on point A, 1 point on line AB and one point on line AC = $1 \times {}^mC_1 \times {}^nC_1$

Total no. of Triangle = $^mC_2 \times {}^nC_1 + {}^mC_1 \times {}^nC_2 + 1 \times {}^mC_1 \times {}^nC_1$

Problem-2 If you have a 10 different chocolates & 20 different Toffees. In how many ways 3 chocolate & 4 toffees can be given to a child.

Ans

Total number of diff. chocolates = 10

" " " " Toffees = 20

The total ways to choose 3 chocolate from the 10 diff chocolate = $^{10}C_3$

$$= \frac{10!}{3!(10-3)!}$$

$$= \frac{10 \times 9 \times 8 \times 7!}{3! \times 7!} = \frac{10 \times 9 \times 8}{3 \times 2} = 120 //$$

The total ways to choose 4 Toffees from the diff. chocolate = $^{20}C_4$

$$= \frac{20!}{4!(20-4)!} = \frac{20 \times 19 \times 18 \times 17 \times 16!}{4 \times 3 \times 2 \times 1 \times 16!}$$

$$= 51 \times 95$$

$$= 4845 //$$

total ways = 120×4845

$= 581400 //$

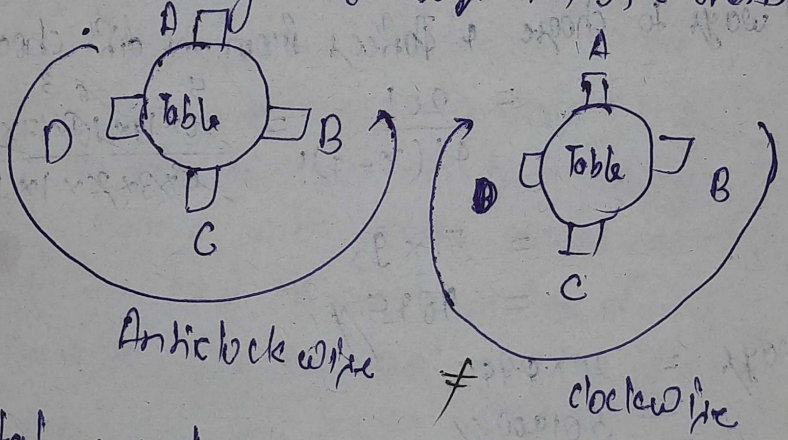
Circular Permutation

arrangment of objects on a circle is know as circular permutation.

If we have n-different objects. Then number of circular permutations will be $(n-1)!$ When anticlockwise and clockwise arrangment are different otherwise $\frac{(n-1)!}{2}$ when anticlockwise & clockwise arrangement are same.

Problem ⇒ In how many ways 4 boys can be arrangement on a circular table.

Ans We are having four boys A, B, C and D



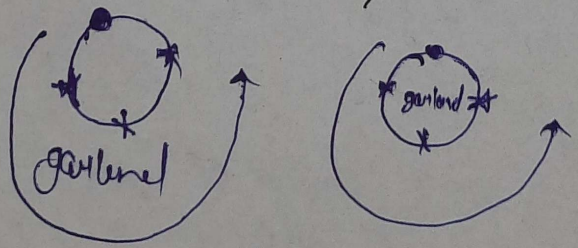
total way to arrange 4 boys in circular Table $= (4-1)!$

$= 3!$

$= 3 \times 2 \times 1 = 6$

Problem ⇒ How many garland can be made by using 4 different flowers.

Ans



It is not change because garland can be rotate

So, Total no. of garland made by 4 diff flowers $= \frac{(4-1)!}{2} = \frac{3\times 2}{2}$

$= 3$

Permutation & combination of

Permutation with repetation

Problem ⇒ 1 Consider 20 objects

A, A, A, A, B, B, B, C, C, C, D, D, E, E, f, G, H, I, Z

In how many ways we can arrange these 20 objects in a line.

Ans The total posibility or ways to arrange in line $= \frac{20!}{5! \cdot 3! \cdot 3! \cdot 2! \cdot 2!}$

Problem ⇒ 2 How many words can be formed by using all the letters of the word "HONOLULU".

A given word-

"HONOLULU"

Total ~~two~~ letter in word "HONOLULU" = 8

The no. of word which formed by using letter of word "HONOLULU" $= \frac{8!}{2! \cdot 2! \cdot 2!}$ //

~~Permutation of Alike objects~~

Combination of Alike objects

Problem consider some objects A, A, B, B, B, C

(i) In How many ways we can select one object.

(ii) " " " " " " " two object.

(iii) " " " " " " " three object.

Ans (i) The total way to select one object from the given objects = 3

(ii) " " " " " two " " " " "

(a) when both object different $= {}^3C_2$

(b) when both object same $= {}^3C_1$

$= {}^3C_2 + {}^3C_1 = 6$

(iii) The total ways to select three objects -

(a) when three objects are same -

$= 1$

(b) when three object are different -

$= {}^3C_3 = \frac{3!}{3!\,(3-3)!} = 1$

(c) when 1 object diff, 2 alike -

$= {}^3C_1 \times {}^2C_1 = \frac{3!}{1!\,(3-1)!} \times \frac{2!}{(2-1)!} = \frac{3 \times 2! \times 1}{2! \times}$

$= 6$

total ways $= 1 + 1 + 6$

$= 8$ //

Problem ⇒ find total number of ways of selection of 4 distinct red balls, 2 identical black balls, one & 3 identical white balls.

Sol^n let 4 distinct red balls, A, B, C, D

2 identical black balls, E, E

3 identical white balls, W, W, W,